

Report metrics
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Metrics comparison

Table 1: Overview of Metrics for Evaluating Synthetic Data

Category	Metric	What discard	Package name
Statistical	Jensen-Shannon Distance		stats.jensenshannon.dist.marginal
	χ^2 Test		stats.chi_squared_test.marginal
	Feature Correlation		
	Inverse KL Divergence		stats.inv_kl_divergence.marginal
	KS Test (Kolmogorov-Smirnov)		stats.ks_test.marginal
	Max Mean Discrepancy (MMD)		stats.max_mean_discrepancy.joint
	Wasserstein Distance		stats.wasserstein_dist.joint
Sanity	PRDC		stats.prdc.coverage, stats.prdc.density, stats.prdc.precision, stats.prdc.recall
	Alpha Precision		stats.alpha_precision.authenticity_naive, stats.alpha_precision.authenticity_OC
	Survival KM Distance		
	Data Mismatch		sanity.data_mismatch.score
	Common Rows Proportion		sanity.common_rows_proportion.score
	Nearest Synthetic Neighbor Distance		sanity.nearest_syn_neighbor_distance.mean
	Close Values Probability		sanity.close_values_probability.score
	Distant Values Probability		
Privacy	Delta-Presence		privacy.delta-presence.score
	k-Anonymity		privacy.k-anonymization.syn
	k-Map		privacy.k-map.score
	l-Diversity		privacy.distinct l-diversity.syn
	Identifiability Score	Score and OC_score	privacy.identifiability_score.score, privacy.identifiability_score.score_OC

Legend:

- : suggest to delete;
- : attention need;
- : okay.

The synthcity python package implement Alpha Precision in statistical measures, not in sanity. For the Alpha Precision metric there are several metrics:

- `stats.alpha_precision.delta_coverage_beta_OC` ;
- `stats.alpha_precision.delta_coverage_beta_naive` ;
- `stats.alpha_precision.delta_precision_alpha_OC` ;
- `stats.alpha_precision.delta_precision_alpha_naive` .

They are used to evaluate the authenticity score, so we can think to maintain only the authenticity one.

Also the `PRDC` is implemented in stats measures.

The χ^2 Test with its `stats.chi_squared_test.marginal` implementation is listed in our metrics but it performs worst in every field. We can think to remove it.

The `KS Test (Kolmogorov-Smirnov)` is implemented in the package, but it performs in almost every case worse. We can think to remove it.

For both `Max Mean Discrepancy (MMD)` and `Wasserstein Distance`, same consideration above.